

Residual Reinforcement Learning for Adaptive Grasp Recovery

Under Dynamics and Grasp Uncertainty

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CS5478 Intelligent Robots: Algorithms and Systems

Proposal Presentation



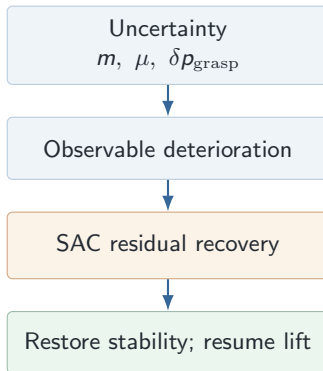
1. Motivation: When a Good Grasp Goes Bad

A Cartesian controller handles reaching, grasping, and lifting **well** — under nominal conditions.

But an initially successful grasp can **deteriorate during lifting** because of:

- ▶ uncertain object mass m ,
- ▶ uncertain friction μ ,
- ▶ imperfect grasp placement δp_{grasp} .

Failure modes: **slip**, **lateral drift**, **rotation**.



Question: how should the robot **recover** once a grasp has begun to deteriorate?

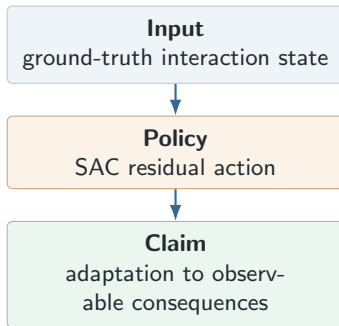
2. Scope: Recovery, Not Estimation

Research question

How can a recovery-specific SAC policy adapt its corrective actions to different grasp-deterioration states under uncertainty in m , μ , and δp_{grasp} ?

What SAC is not asked to do

- ▶ estimate m or μ ,
- ▶ learn perception or slip detection,
- ▶ plan regrasps after a full drop.



SAC responds to **observable consequences**,
not to hidden parameters.

Conventional control does the task. RL is responsible only for **how to recover**.

3. Task Setup and Uncertainty Model

Robot and task

- ▶ 7-DoF Panda arm in MuJoCo,
- ▶ parallel-jaw gripper,
- ▶ approach $\rightarrow$ descend $\rightarrow$ close $\rightarrow$ lift,
- ▶ **one fixed cylindrical geometry.**

Controlled grasp placement

$$p_{\text{grasp}} = p_{\text{ideal}} + \delta p_{\text{grasp}}, \quad \delta p_{\text{grasp}} = [\delta x, \delta y, \delta z].$$

Mass
 m randomized

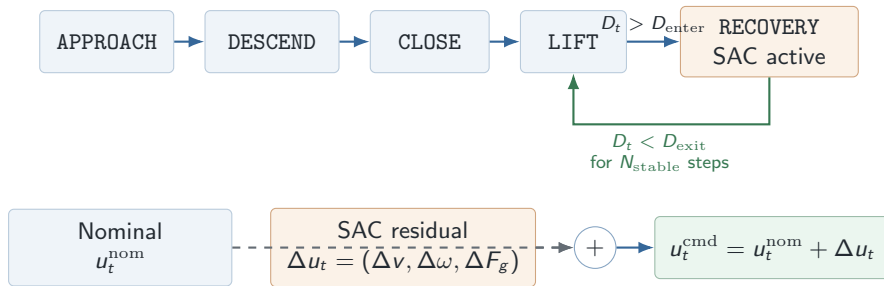
Friction
 μ randomized

Placement
 δp_{grasp} randomized

Geometry fixed
controlled study

Training uses sampled combinations; evaluation reserves **held-out regions** of the joint uncertainty space.

4. Mode-Switched Recovery Architecture



$D_{\text{exit}} < D_{\text{enter}}$ (hysteresis: prevents rapid mode switching).

RL intervenes **only** when the grasp is deteriorating; otherwise the nominal controller runs untouched.

5. Deterioration Detection

Reference: stable post-grasp relative pose. With

$$p_{\text{rel},t} = p_{\text{object},t} - p_{\text{gripper},t}, \quad v_{\text{rel},t} = v_{\text{object},t} - v_{\text{gripper},t},$$

$$D_t = w_p \frac{\|\Delta p_{\text{rel},t}\|}{p_{\text{scale}}} + w_v \frac{\|v_{\text{rel},t}\|}{v_{\text{scale}}} + w_\theta \frac{|\theta_{\text{rel},t}|}{\theta_{\text{scale}}} \\ + w_\omega \frac{\|\omega_{\text{rel},t}\|}{\omega_{\text{scale}}} + w_c C_t,$$

with

$$C_t = \left| \frac{f_{L,t} - f_{R,t}}{f_{L,t} + f_{R,t} + \epsilon} \right| + \mathbf{1}[\text{contact loss}].$$

The detector is a **shared experimental instrument**, not a contribution.

Read-out

- ▶ D_t : severity of deterioration,
- ▶ $\dot{D}_t$: improving or worsening,
- ▶ all from MuJoCo ground truth.

Why it is fixed

- ▶ removes perception as a variable,
- ▶ identical trigger for every method,
- ▶ isolates **recovery** as the object of study.

6. SAC Recovery Controller

Residual action

$$a_t = [\Delta v_x, \Delta v_y, \Delta v_z, \Delta \omega_x, \Delta \omega_y, \Delta \omega_z, \Delta F_g].$$

$$\begin{bmatrix} v_t^{\text{cmd}} \\ \omega_t^{\text{cmd}} \end{bmatrix} = \begin{bmatrix} v_t^{\text{nom}} \\ \omega_t^{\text{nom}} \end{bmatrix} + \begin{bmatrix} \Delta v_t \\ \Delta \omega_t \end{bmatrix}, \quad F_{g,t}^{\text{cmd}} = F_{g,t}^{\text{nom}} + \Delta F_{g,t}.$$

If $\Delta v_z < -v_z^{\text{nom}}$, SAC can **reverse the lift** — not only “squeeze harder.”

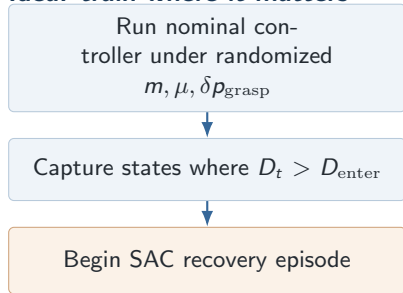
Observations

- ▶ $q, \dot{q}$, end-effector state,
- ▶ object pose and motion,
- ▶ relative state, contact forces,
- ▶ nominal command, $D_t, \dot{D}_t$.

m and μ are **deliberately excluded**.

7. Recovery-Focused Training

Idea: train where it matters



Initial states are **stratified by** D_t into mild / moderate / severe. A curriculum is an *optional* extension, not on the critical path.

Objective: **minimum intervention** sufficient for recovery — but strong action is allowed when it prevents failure.

Potential-based shaping

$$\Phi(s_t) = -\lambda_D D_t, \quad r_t^{\text{shape}} = \gamma \Phi(s_{t+1}) - \Phi(s_t).$$

$$\begin{aligned} r_t = & r_t^{\text{shape}} - \lambda_f F_{g,t}^2 - \lambda_c (\|\Delta v_t\|^2 + \|\Delta \omega_t\|^2) \\ & - \lambda_t + \mathbf{1}[\text{success}] r_{\text{rec}} \\ & - \mathbf{1}[\text{drop}] r_{\text{drop}}. \end{aligned}$$

8. Staged Development and Baselines

Staged action space

Stage A — low-dimensional

$$a_t = [\Delta v_z, \Delta F_g].$$

Lift / grip-force trade-off.

Stage B — Cartesian (conditional)

If offsets cause lateral / rotational failures:

$$a_t = [\Delta v, \Delta \omega, \Delta F_g].$$

Three controllers

1. **Nominal.** No recovery mechanism.
2. **Hand-designed reactive.** Same detector, same residual bounds:

$$\Delta F_g = k_{p,F} D_t + k_{d,F} \dot{D}_t,$$

$$\Delta v_z = -\text{clip}(k_{p,v} D_t + k_{d,v} \dot{D}_t, 0, \Delta v_{z,\max}).$$

3. **SAC.** Same detector, same information, learned action.

The heuristic and SAC share bounds, limits, detector, and information — **only the action-selection strategy differs.**

9. Experimental Evaluation

Primary metric

$$\text{Recovery Rate} = \frac{N_{\text{deteriorated grasps stabilized}}}{N_{\text{recovery attempts}}}.$$

Additional metrics

- ▶ drop rate, maximum relative slip,
- ▶ time-to-recovery,
- ▶ gripping effort, residual magnitude,
- ▶ subsequent task-completion rate,
- ▶ confidence intervals over episodes.

Comparisons

- ▶ nominal vs. heuristic vs. SAC,
- ▶ mild / moderate / severe initial D_t ,
- ▶ held-out $m-\mu$ combinations,
- ▶ increasing δp_{grasp} ,
- ▶ 2D vs. Cartesian residual,
- ▶ SAC with / without contact features.

Behavior analysis

Trajectories of $D_t, F_g, v_z, \Delta v_z$ per severity bin; failure taxonomy (insufficient / destabilizing / unrecoverable).

10. Expected Contribution and Scope

Contribution

- ▶ recovery-focused RL training on rare deteriorating states,
- ▶ controlled comparison against a *tuned* reactive controller,
- ▶ characterization of when learned recovery helps,
- ▶ evaluation on held-out regions of uncertainty.

The result does **not** depend on SAC universally beating the heuristic; characterizing *where* learning helps is part of the study.

Core scope

- ▶ one cylinder geometry,
- ▶ nominal Cartesian controller,
- ▶ mass / friction / offset uncertainty,
- ▶ ground-truth detector,
- ▶ recovery-focused SAC,
- ▶ nominal + reactive baselines.

Outside scope

Perception uncertainty, system identification, latent-state learning, cross-object generalization, RGB-D planning, post-drop regrasping.

11. Implementation Plan and Risk Control

Minimum complete project

1. Stable nominal grasp-and-lift controller
2. Randomized m , μ , and δp_{grasp}
3. Calibrated ground-truth deterioration detector
4. Stage A SAC: $[\Delta v_z, \Delta F_g]$
5. Tuned reactive baseline
6. Held-out uncertainty evaluation

Extensions only if justified

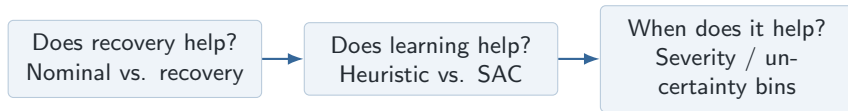
- ▶ full Cartesian residual action,
- ▶ curriculum for severe recovery states,
- ▶ contact-feedback ablation.

Risk control

If richer actions do not improve recovery, the project remains complete with the low-dimensional controller. If SAC does not outperform the heuristic, the comparison still identifies regimes where learning is or is not useful.

Protect the **controlled recovery comparison** first; treat added action-space complexity as an extension.

12. What the Experiments Will Tell Us



Recovery value

Recovery rate and drop rate quantify whether intervening after deterioration is worthwhile.

Policy value

Matched detector, observations, and action bounds isolate learned vs. fixed action selection.

Behavior insight

D_t , F_g , v_z , and residual trajectories reveal how intervention changes with severity.

The central result is not a single win rate; it is a **map of recovery behavior across deterioration regimes**.

Discussion

Nominal manipulation → detect deterioration → SAC recovery → resume task

Focused question: how should the robot recover once a grasp has begun to deteriorate?